

Anti-virus & Virus Technologies

Part III – Anti-Virus Programs



YEAR 1 – SEMESTER 1

Bucharest 2010

Anti-virus & Virus Technologies



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YEAR 1 – SEMESTER 1

Bucharest 2010

7 Virus definitions

- *"A virus is a program that is able to infect other programs by modifying them to include a possibly evolved copy of itself"* (Dr. Frederick B. Cohen , A Short Course on Computer Viruses)
- *"A computer virus is a program that recursively and explicitly copies a possibly evolved version of itself"* (Peter Szor, The Art of Computer Virus Research and Defense)

7 Virus definitions

- **Worms** – network viruses which replicate on networks
 - Mailers and Mass-Mailer Worms – worms that replicate by email
 - Octopus – complex worm composed from multiple programs placed in different locations on a network
 - Rabbits – worm that “jumps” itself on different network hosts

7 Virus definitions

- **Logic Bombs** - a programmed malfunction of a legitimate application that will execute when some conditions are met
- **Trojan Horses** - malicious program that provide different services to the attacker
 - Backdoors (Trapdoors) – provides remote access
 - Password-Stealing Trojans – records and send passwords

7 Virus definitions

- Exploits – specific to a single or set of vulnerabilities
- Downloaders – malicious program that downloads and executes other programs
- Dialers – calls to premium numbers without notifying users
- Droppers (installers) – first generation virus code

7 Virus definitions

- Injectors – special droppers that install the virus in memory
- Auto-Rooters – collection of exploits used to gain root rights on a target machine
- Kits (virus generation) – Virus Creation Laboratory (VCL) used to generate different versions

7 Virus definitions

- Flooders – used for Denial of Service attacks (DoS)
- Keyloggers – record and transmit all typed keys
- Rootkits – modified versions of common tools



7 Virus definitions

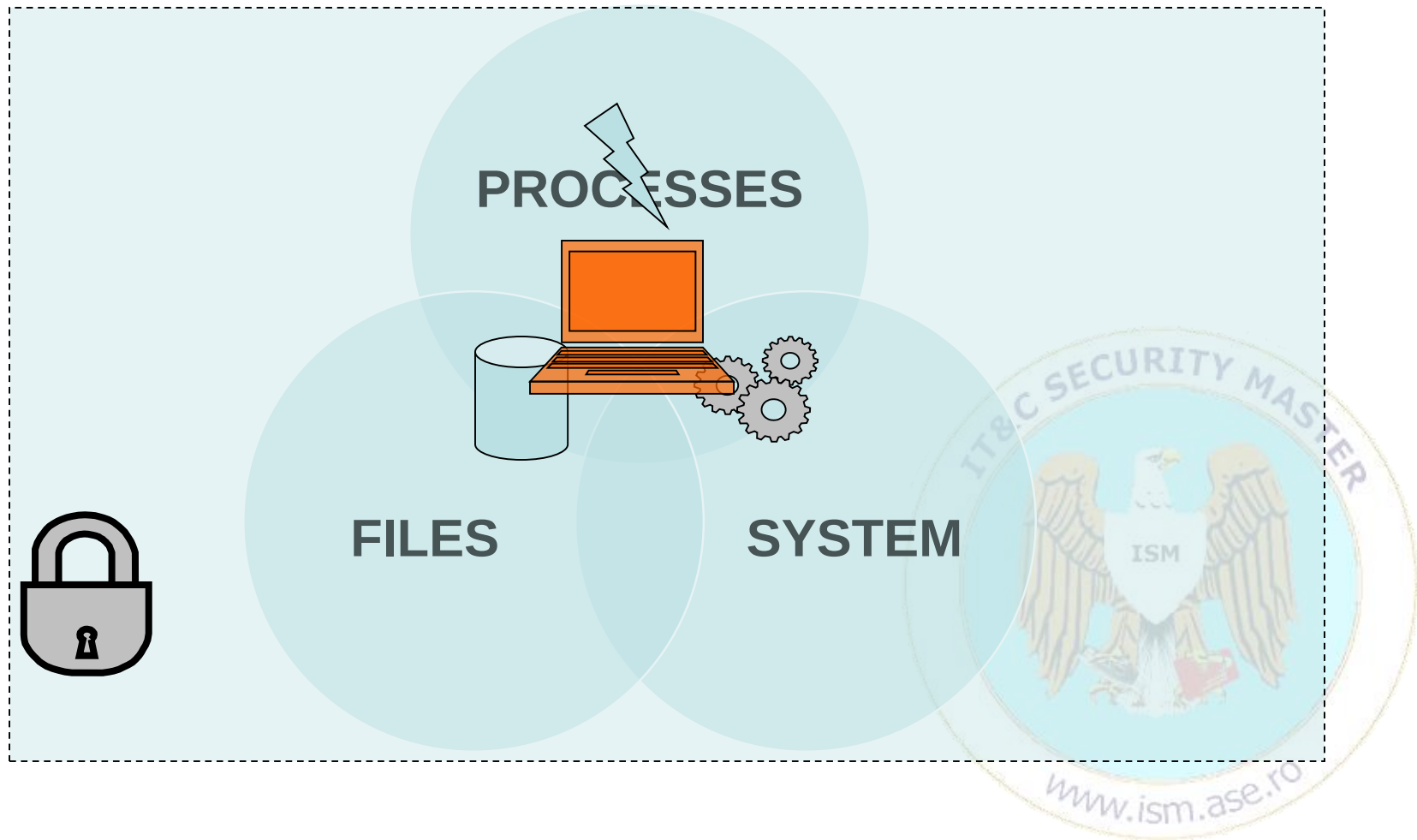
- Exploits – attacks specific to a single vulnerability or set of vulnerabilities
- Downloaders
- Dialers
- Droppers
- Injectors



1011110110101010
0110101110001011
1010011001010011

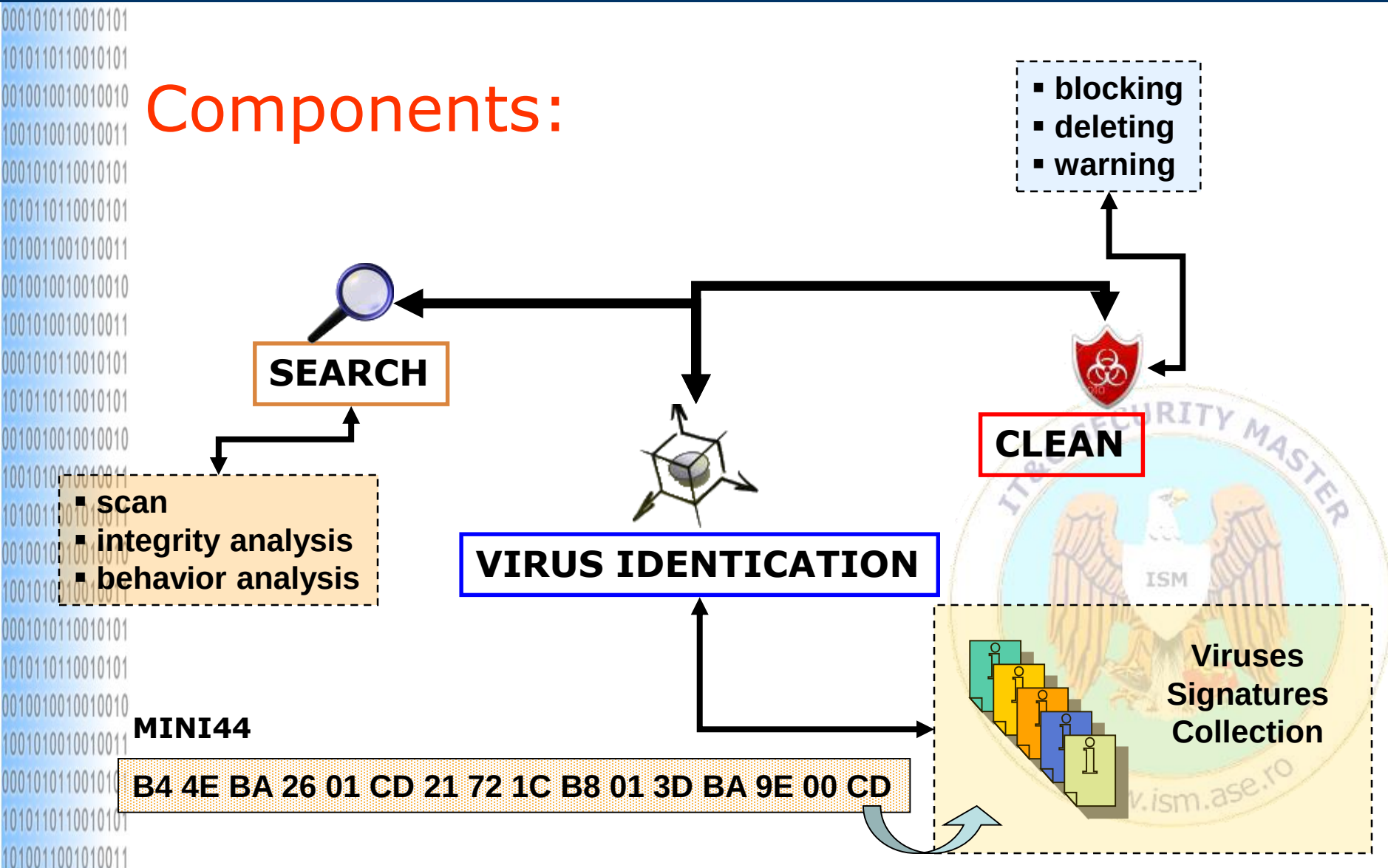
7 Anti-Virus Programs

0001010110010101
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1010011001010011
0010010010010010
1001010010010011
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7 Anti-Virus Programs

Components:



7 Anti-Virus Programs

Advanced Scanning Methods:

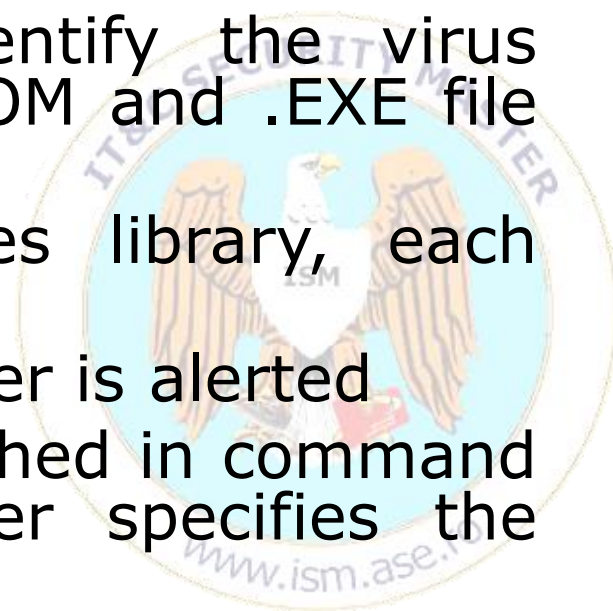
- Wildcards
- Generic degree
- Top-and-Tail scanning
- Entry-Point & Fixed-Point scanning
- Hashing
- Smart scanning
- Skeleton detection



7 Anti-Virus Programs

AV-SCAN:

- Antivirus program which contains the scanning routines for RAM and HDD
- Scans the RAM in order to identify the memory resident virus programs
- Scans HDD in order to identify the virus programs which infect the .COM and .EXE file host programs
- Contains a viruses signatures library, each signature using 16 bytes
- In case of infection the end-user is alerted
- The antivirus program is launched in command line – optionally a parameter specifies the analyzed HDD drive letter

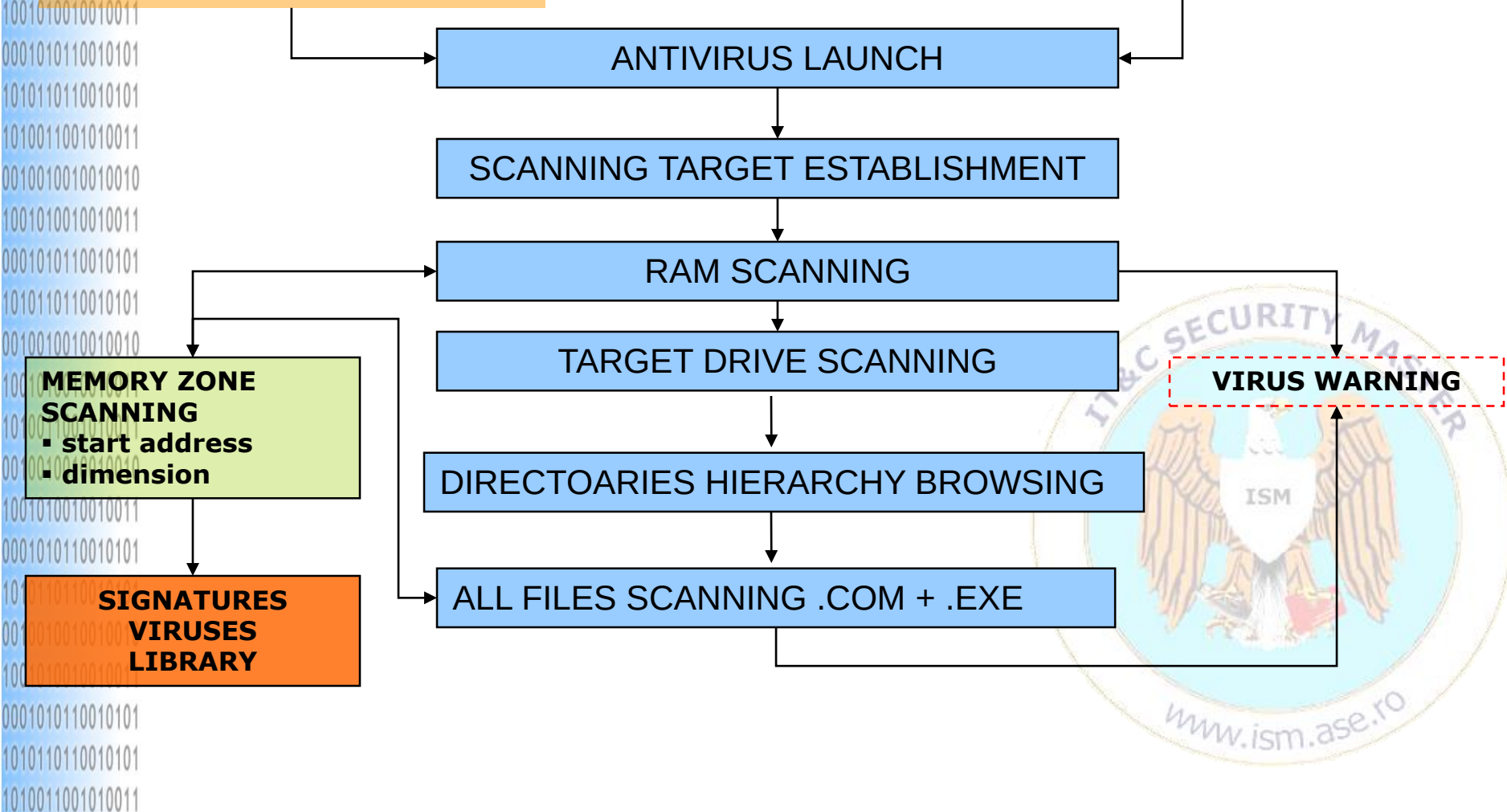


7 Anti-Virus Programs

AV-SCAN

parameter: the drive for scanning
d:\templavscan.com c:

the default parameter is the current drive letter
d:\templavscan.com



7 Anti-Virus Programs

AV-SCAN

Memory Variables:

DBUF_SIZE equ 16384

; the dimension for the scanning buffer scanat

EXEFLAG equ 00000100B

; the flags used for indicating the scanning scope

COMFLAG equ 00001000B

; check EXE file

RAMFLAG equ 00010000B

; check COM file

ENDLIST equ 00100000B

; RAM checking

; check the end of the flags list

INFECTED DB ' It is infected by the virus \$'

; warning message

VIRUS_ST DB ' virus.',0DH,0AH,24H

RAM_NAME DB 7,7,7,7,7,'ACTIVE MEMORY',0

; warning message for RAM

EXE_FILE DB '*.EXE',0

; EXE file mask

COM_FILE DB '*.COM',0

; COM file mask

ANY_FILE DB '*.*',0

; all files + directories

ROOT DB "\",0

**; command string cd **

UP_ONE DB '..',0

; command string cd..

NAME_STRINGS DB 'MINI-44','\$'

; the viruses name

DB 'Kilroy-B dropper\$'

DB 'Yellow Worm\$'

PATH db 800 dup(1)

; the current PATH for scanning

CURR_DIR db 64 dup(2)

; the current directory

DSIZE dw ?

; the dimension for the scanning area

SEARCH_REC db 43 dup(3)

; DTA – for the searching – SEARCH results

CURR_DR DB ?

; the current disk drive

DISK_DR db ?

; the scanned drive

FFLAGS db ?

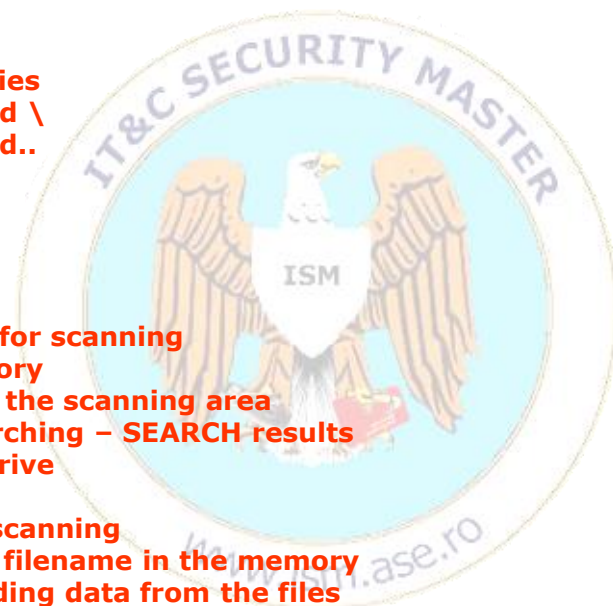
; the flags used in scanning

FILE_NAME dw ?

; the pointer to the filename in the memory

DATA_BUF DB DBUF_SIZE DUP (?)

; the buffer for reading data from the files



7 Anti-Virus Programs

AVSCAN:

```
mov ax,cs
mov ds,ax
```

```
mov ah,19h
int 21h
mov byte ptr [CURR_DR],al
```

```
mov ah,47h
mov dl,0
mov si,offset CURR_DIR
int 21h
```

```
mov bx,5ch
mov al,es:[bx]
or al,al
jnz AVS1
mov ah,19h
int 21h
inc al
```

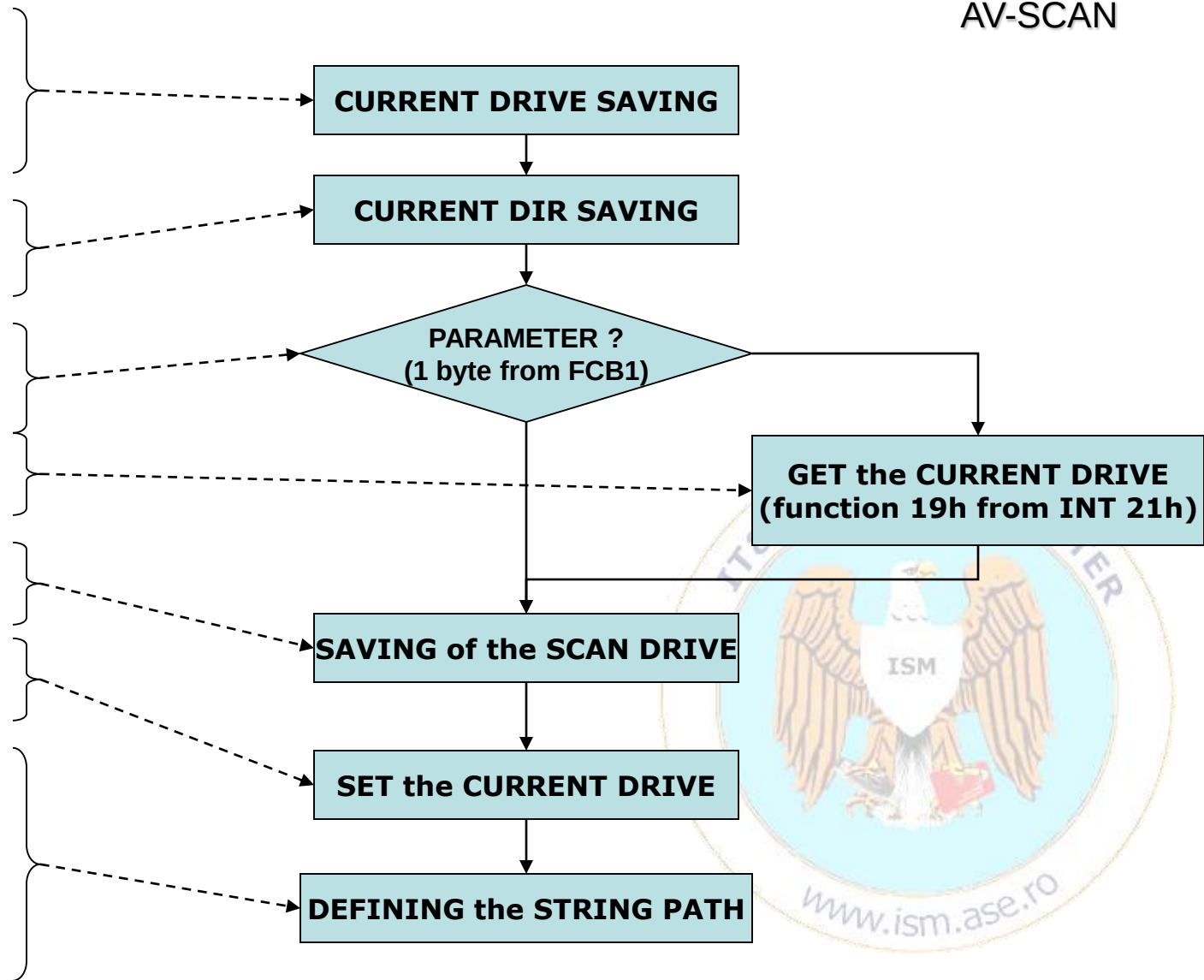
AVS1:

```
dec al
mov byte ptr [DISK_DR],al
```

```
mov dl,al
mov ah,0eh
int 21h
```

```
push cs
pop es
mov di,offset PATH
mov al,[DISK_DR]
add al,'A'
mov ah,':'
stosw
mov ax,'\\'
stosw
```

AV-SCAN



7 Anti-Virus Programs

AV-SCAN

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1001010010010011

0001010110010101

1010110110010101

0010010010010010

1001010010010011

0001010110010101

1010110110010101

1010011001010011

```
mov dx, offset HELLO
mov ah, 09h
int 21h
```

```
call SCAN_RAM
jc AVS4
```

AVS3:

```
mov dx, offset ROOT
mov ah, 3bh
int 21h
```

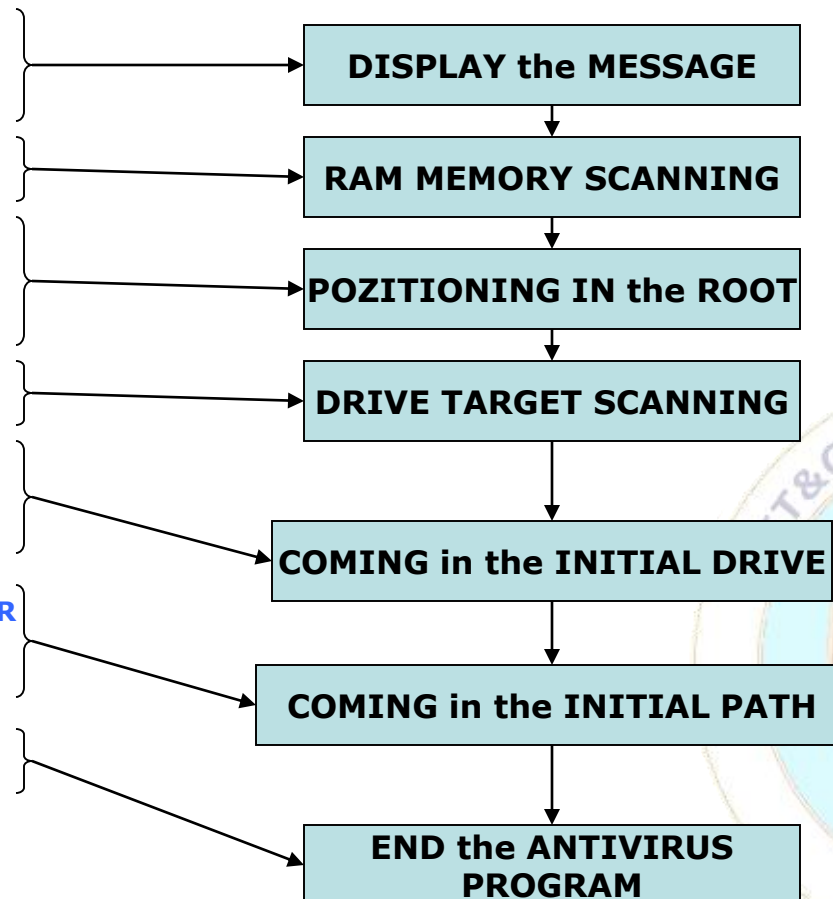
```
call SCAN_ALL_FILES
```

AVS4:

```
mov di, [CURR_DR]
mov ah, 0eh
int 21h
```

```
mov dx, offset CURR_DIR
mov ah, 3bh
int 21h
```

```
mov ax, 4C00h
int 21h
```



7 Anti-Virus Programs

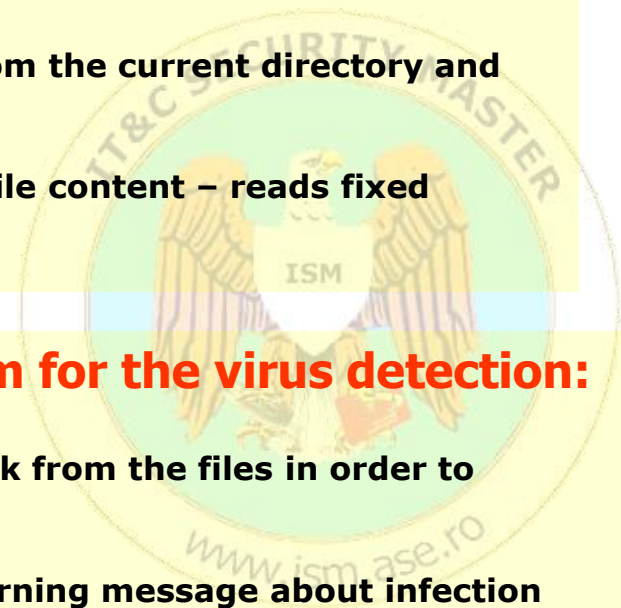
AV-SCAN

The routines used by the anti-virus program for SCANNING:

- **SCAN_RAM** – tests the memory for TSR virus identification
- **SCAN_ALL_FILES** – recursive procedure used for the directories browsing from the target drive + COM & EXE file identification
- **SCAN_COM** – the procedure identifies COM files from the current directory and calls the content scanning procedure
- **SCAN_EXE** – the procedure identifies the EXE files from the current directory and calls the content scanning routine
- **SCAN_FILE** – the analysis procedure for COM & EXE file content – reads fixed amount of bytes from the identified files

The routines used by the anti-virus program for the virus detection:

- **SCAN_DATA** – the procedure tests the read bytes block from the files in order to identify the virus signature
- **DISP_VIRUS_NAME** – the procedure displays the warning message about infection



7 Anti-Virus Programs

AV-SCAN



SCANNING PROCEDURE

SCAN_RAM

SCAN_ALL_FILES

SCAN_EXE

SCAN_COM

The current directory
analysis

SCAN_FILE

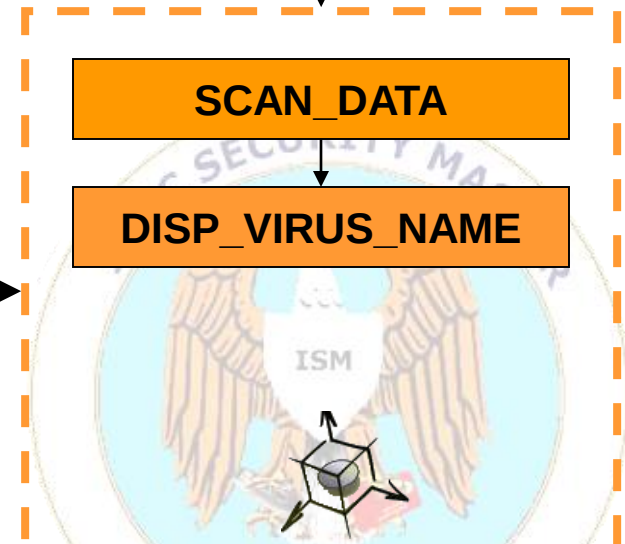
Recursive call for browsing
all the directories

SCAN_DATA

DISP_VIRUS_NAME

ISM

VIRUS DETECTION Procedure



7 Anti-Virus Programs

AV-SCAN

SCAN_RAM:

```
mov word ptr[FILE_NAME],offset RAM_NAME
xor ax,ax
mov es,ax
mov bx,ax
```

;set ES:BX = 0

SRL:

```
mov ah,RAMFLAG
mov cx,800h
call SCAN_DATA
pushf
mov ax,es
add ax,800h
mov es,ax
popf
jc SREX
or ax,ax
jnz SRL
clc
```

;the scanning block dimension
;call check routine

;modifies ES for the next bytes chunk

;EXIt if a virus exist
;check the RAM scanning status – finished?

;CLEAN MEMORY

SREX:

```
ret
```

;Scanning procedure
;for RAM < 1 MB
;with TSR viruses
;if a virus is found
;then sets CF=1



7 Anti-Virus Programs

AV-SCAN

WHY do we need 2 DTAs with 43 bytes each? – Directories/Sub-directories and respectively files?



7 Anti-Virus Programs

SCAN_ALL_FILES:

```
push bp
mov bp,sp
sub bp,43
mov sp,bp
```

```
mov dx,offset SEARCH_REC
mov ah,1ah
int 21h
```

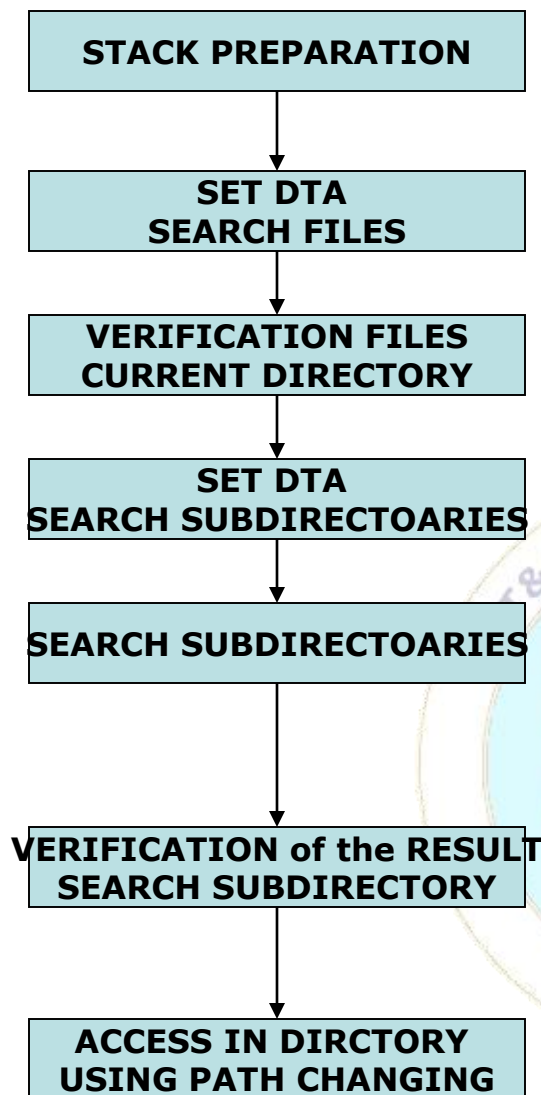
```
call SCAN_COM
call SCAN_EXE
```

```
mov dx,bp
mov ah,1AH
int 21H
```

```
mov dx,OFFSET ANY_FILE
mov ah,4EH
mov cx,10H
int 21H
```

SAFLP:

```
or al,al
jnz SAFEX
cmp BYTE PTR [bp+1Eh],','
je SAF1
test BYTE PTR [bp+15h],10H
jz SAF1
lea dx,[bp+1Eh]
mov ah,3Bh
int 21H
```



AV-SCAN

Reserve 43 bytes on the stack necessary for the SEARCH

SCAN_COM & SCAN_EXE procedures use the searching methods

For directory searching the DTA is on the procedure stack

Use function 4Eh with the parameter CX = 10h and the mask *.*

Avoid the hidden directories or the entries that are not directories – the 21th byte from DTA shows the type

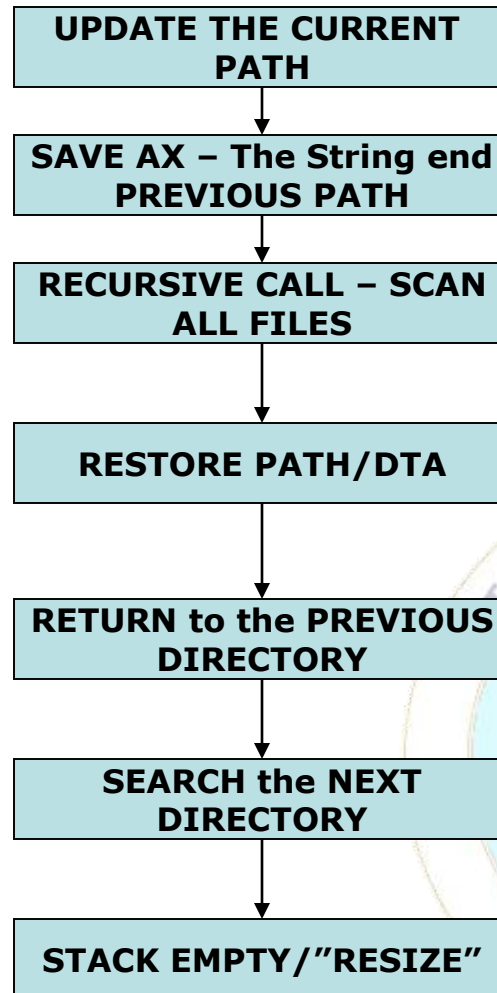
Positioning in the new directory – chdir/cd

7 Anti-Virus Programs

AV-SCAN

SCAN_ALL_FILES:

```
call UPDATE_PATH  
push ax  
  
call SCAN_ALL_FILES  
  
pop bx  
mov BYTE PTR [bx],0  
mov dx,bp  
mov ah,1Ah  
int 21H  
  
mov dx,OFFSET UP_ONE  
mov ah,3BH  
int 21H  
SAF1:  
mov ah,4FH  
int 21H  
jmp SAFLP  
SAFEX:  
add bp,43  
mov sp,bp  
pop bp  
ret
```



PATH String update

AX has the offset of the end for the char string – THE PATH

refacere string PATH la revenire din apel recursiv

Return to the parent directory *cd..*

Next the sub-directory

Repositioning SP

7 Anti-Virus Programs

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0001010110010101

1010110110010101

0010010010010010

1001010010010011

1010011001010011

0010010010010010

1001010010010011

0001010110010101

1010110110010101

0010010010010010

1001010010010011

0001010110010101

1010110110010101

1010011001010011

SCAN_EXE:

```

mov byte ptr[FFLAGS],EXEFLAG and 255
mov word ptr[FILE_NAME],offset SEARCH_REC + 1Eh
mov dx,offset EXE_FILE
jmp SCAN_FILES
  
```

SCAN_COM:

```

mov byte ptr[FFLAGS],COMFLAG
mov word ptr[FILE_NAME],offset SEARCH_REC + 1Eh

mov dx,offset COM_FILE
  
```

SCAN_FILES:

```

mov ah,4eh
mov cx,3fh ; fisiere executabile?
int 21h
  
```

SCLP:

```

or al,al
jnz SCDONE
call SCAN_FILE
  
```

```

mov ah,4fh
int 21h
jmp SCLP
  
```

SCDONE:

```

ret
  
```

AV-SCAN

SET the POINTER to the
FILENAME from DTA

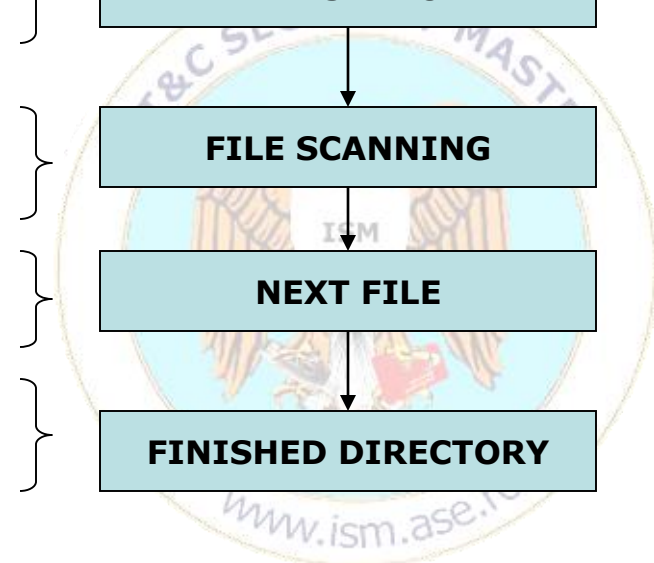
SET the POINTER to the
FILENAME from DTA

FILE SEARCH

FILE SCANNING

NEXT FILE

FINISHED DIRECTORY



7 Anti-Virus Programs

AV-SCAN

SCAN_FILE:

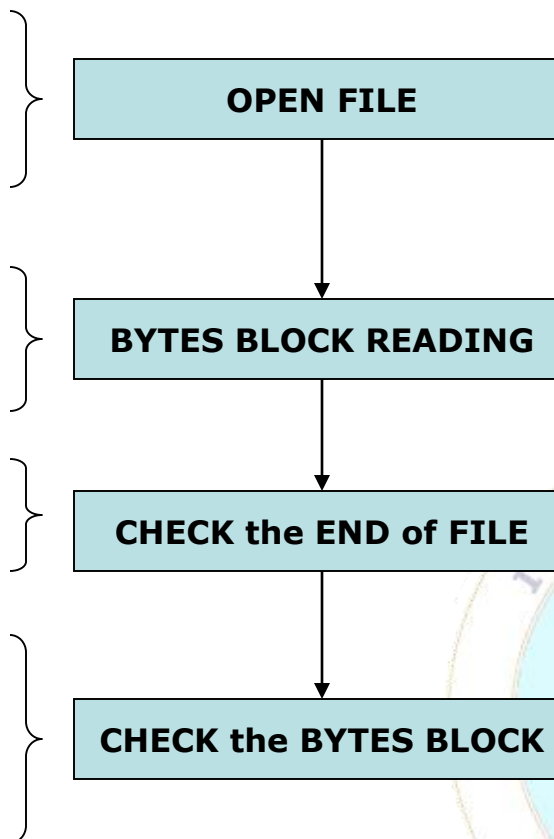
```
mov dx,word ptr[FILE_NAME]
mov ax,3d00h
int 21h
jc SFCLOSE
mov bx,ax
```

SF1:

```
mov ah,3fh
mov cx,DBUF_SIZE
mov dx,offset DATA_BUF
int 21h
```

```
cmp ax,16
jle SFCLOSE
```

```
mov cx,ax
push bx
mov bx,offset DATA_BUF
push ds
pop es
mov ah,[FFLAGS]
call SCAN_DATA
```



7 Anti-Virus Programs

AV-SCAN

SCAN_FILE:

```
pop bx  
jc SFCL2
```

```
mov ax,4201h  
mov cx,-1  
mov dx,-16  
int 21h  
jmp SF1
```

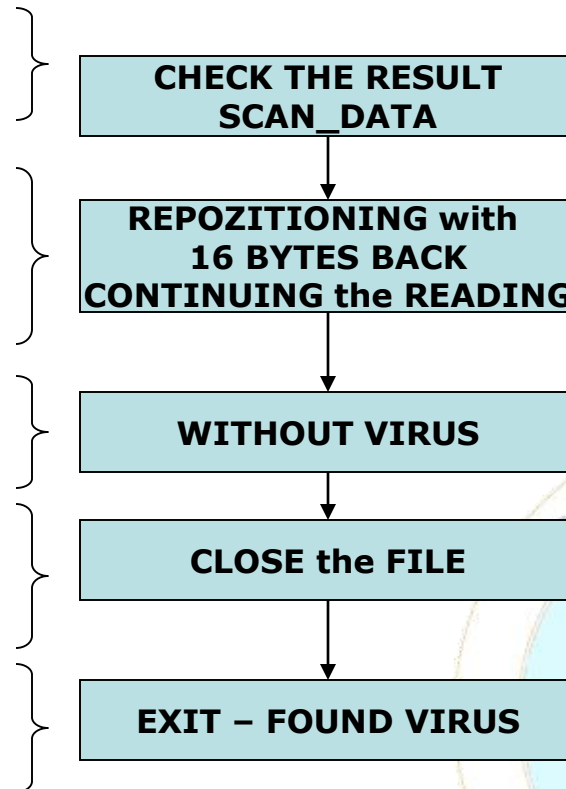
SFCLOSE:

```
clc
```

SFCL2:

```
pushf  
mov ah,3eh  
int 21h  
popf
```

```
ret
```



7 Anti-Virus Programs

AV-SCAN

SCAN_DATA:

- the procedure checks if the bytes array, from the address pointed by ES:BX, contains a virus
- receives the bytes number for checking in CX
- gets in AH the mask for the file type which will be checked
- sets CF flag if a virus is found
- the methods decrypt the virus signature using XOR with AAh value; the encryption is necessary for hiding the anti-virus program library by other similar applications
- the checking of the bytes array is done through byte by byte comparing with the viruses signatures; in order to establish a positive identification, the entire sequence of 16 bytes must be in the tested bytes array
- taking into account the file flag only the proper/certain virus signature are tested

SCAN_DATA:

```
mov word ptr [DSIZE],cx
```

```
mov si,offset SCAN_STRINGS
```

SD1:

```
lodsb
```

```
push ax
```

```
and al,ENDLIST
```

```
pop ax
```

```
jnz SDR
```

```
and al,ah
```

```
jz SDNEXT
```

SET THE BYTES NUMBER
FOR the CHECKING

TESTING THE END OF THE
SIGNATURES VIRUSES LIST

Each signature has 16 bytes and is preceded by a flag byte which indicates of the virus type

7 Anti-Virus Programs

1001010010010011

0001010110010101

1010110110010101

0010010010010010

1001010010010011

0001010110010101

1010110110010101

1010011001010011

0010010010010010

1001010010010011

0001010110010101

1010110110010101

0010010010010010

1001010010010011

1010011001010011

0010010010010010

1001010010010011

0001010110010101

1010110110010101

0010010010010010

1001010010010011

0001010110010101

1010110110010101

1010011001010011

SCAN_DATA:

```
mov dx,bx
add dx,[DSIZE]
mov di,bx
```

SD2:

```
mov al,[si]
xor al,0AAh
```

```
cmp di,dx
je SDNEXT
```

```
cmp al,es:[di]
je SD3
inc di
jmp SD2
```

SD3:

```
push si
push di
mov cx,16
```

SD4:

```
lodsb
xor al,0AAh
inc di
cmp al,es:[di-1]
loopz SD4
```

SET the BEGIN & END
LANDMARKS
for the TESTED BLOCK

DECRYPT the
SIGNATURE BYTE

CHECK the BUFFER END

CHECK the BYTE

IF the BYTE is the same
with the first BYTE FROM
the SIGNATURE
THEN
ALL 16 BYTES ARE CHECKED

AV-SCAN



7 Anti-Virus Programs

AV-SCAN

SCAN_DATA:

```
pop di
pop si
pushf
inc di
popf
```

```
jne SD2
```

```
mov di,si
sub di,offset SCAN_STRINGS+1
mov ax,di
mov di,17
xor dx,dx
div di
mov di,ax
```

```
call DISP_VIRUS_NAME
```

```
stc
ret
```

SDNEXT:

```
add si,16
jmp SD1
```

SDR:

```
clc
ret
```

TEST 16 BYTES BLOCK

DETERMINE the
SIGNATURE NUMBER

DISPLAY the MESSAGE

END - FOUND VIRUS

CHECK the next SIGNATURE

END -
VIRUS NOT FOUND



7 Anti-Virus Programs

UPDATE_PATH:

- The procedure is used for modifying the PATH string
- Saves in DX the offset of the initial string
- Is called by the SCAN_ALL_FILES procedure with browsing the directories structure

AV-SCAN

UPDATE_PATH:

```
lea di,[bp+30]  
mov si,OFFSET PATH
```

```
SAF01:  
lodsb  
or al,al  
jnz SAF01  
dec si
```

```
mov dx,si
```

```
push cs  
pop es  
xchg si,di
```

```
SAF02:  
lodsb  
stosb  
or al,al  
jnz SAF02  
dec di  
mov ax,'\  
stosw  
mov ax,dx  
ret
```

IDENTIFY the end of the
variable PATH

SAVE the end of the STRING

ADD DIRECTORY
+folder_name\



7 Anti-Virus Programs

DISP_VIRUS_NAME:

mov si,OFFSET PATH

FV00:

lodsb

or al,al

jz FV01

mov ah,0EH

int 10H

jmp FV00

FV01:

mov si,[FILE_NAME]

FV02:

lodsb

or al,al

jz FV05

mov ah,0EH

int 10H

jmp FV02

FV05:

mov si,OFFSET NAME_STRINGS

FV1:

or di,di

jz DISP_NAME

push di

FV2:

lodsb

cmp al,'\$'

jnz FV2

pop di

dec di

jmp FV1

AV-SCAN

DISPLAY CHAR BY CHAR
THE PATH

The procedure receives in DI the position of the virus name in list

DISPLAY CHAR BY CHAR
THE FILENAME

DETERMINE
the VIRUS NAME

For DI=0 IT is displayed
the FILENAME

BROWSING THE VIRUSES
NAMES LIST UNTILL
DI = 0



7 Anti-Virus Programs

0001010110010101

1010110110010101

0010010010010010

1001010010010011

0001010110010101

1010110110010101

1010011001010011

0010010010010010

1001010010010011

0001010110010101

1010110110010101

0010010010010010

1001010010010011

1010011001010011

0010010010010010

1001010010010011

0001010110010101

1010110110010101

0010010010010010

1001010010010011

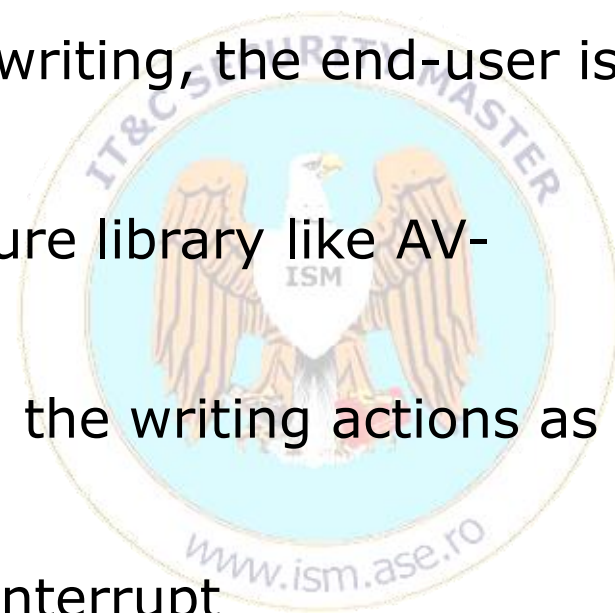
0001010110010101

1010110110010101

1010011001010011

AV-CHECK:

- The antivirus program contains the behavior checking procedures for other applications
- Interrupts all the attempts of the BOOT writing, .COM and .EXE file writing or TSR becoming
- In case of the operation blocking for writing, the end-user is asked to approve/dismiss the action
- AV-CHECK DOESN'T contain a signature library like AV-SCAN
- Has generic approach, considering all the writing actions as being potential viral
- Realizes the *hook* on the 13h & 21h Interrupt



7 Anti-Virus Programs

AV-CHECK

AV-CHECK

d:\templavcheck.com

LAUNCH ANTI-VIRUS

INTRERRUPT 13h HOOK

INTRERRUPT 21h HOOK

EXTERN APPLICATION

INTRERRUPT CALL
13h or 21h

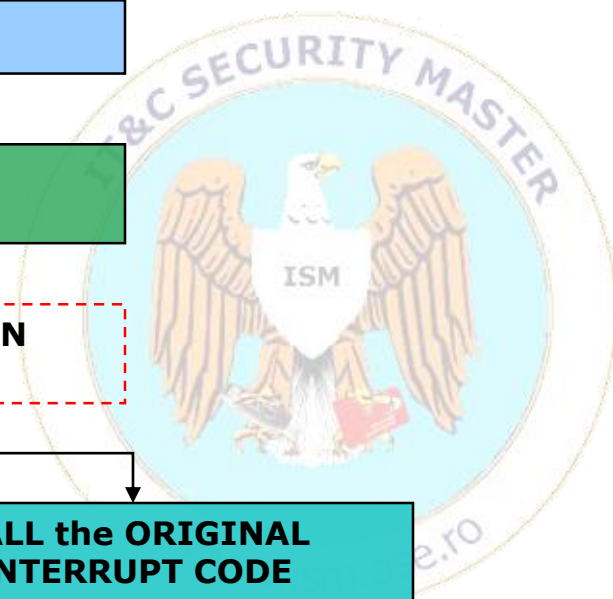
LOAD IN RAM + TSR

CAPTURE the INTERRUPT
13h or 21h

WARNING POSSIBLE VIRUS ACTION
(the end-user decision)

BLOCKING the
ACTION

CALL the ORIGINAL
INTERRUPT CODE



7 Anti-Virus Programs

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1010110110010101
0010010010010010
1001010010010011
0001010110010101
1010110110010101
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0010010010010010
1001010010010011
0001010110010101
1010110110010101
0010010010010010
1001010010010011

- AV-CHECK – Behavior Routines



7 Anti-Virus Programs

TO DO

32 bits EXE + Boot Viruses



Anti-virus & Virus Technologies

THANK YOU!

Cristian Toma

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YEAR 1 – SEMESTER 1

Bucharest 2010